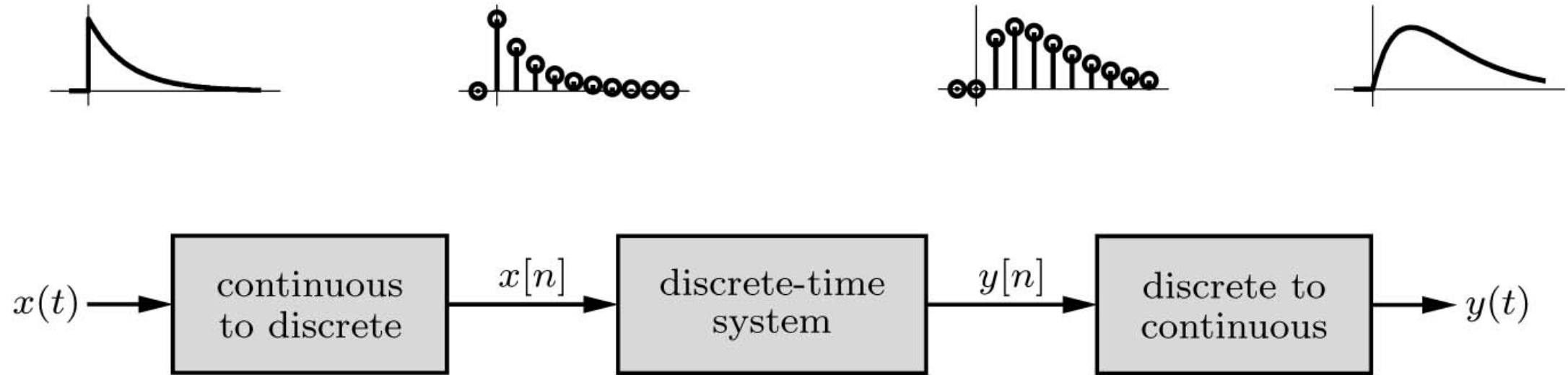


Discrete-Time Signals and Systems

Chapter 4

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University of Central Oklahoma

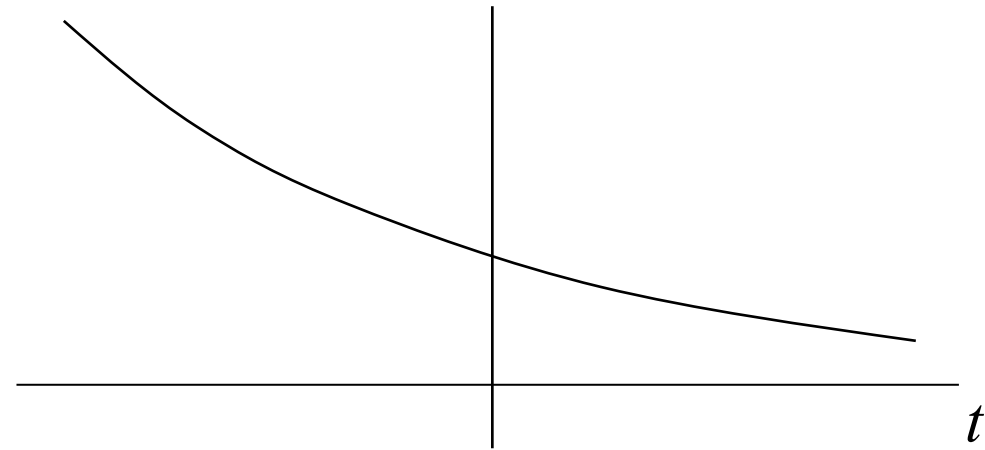
Outline



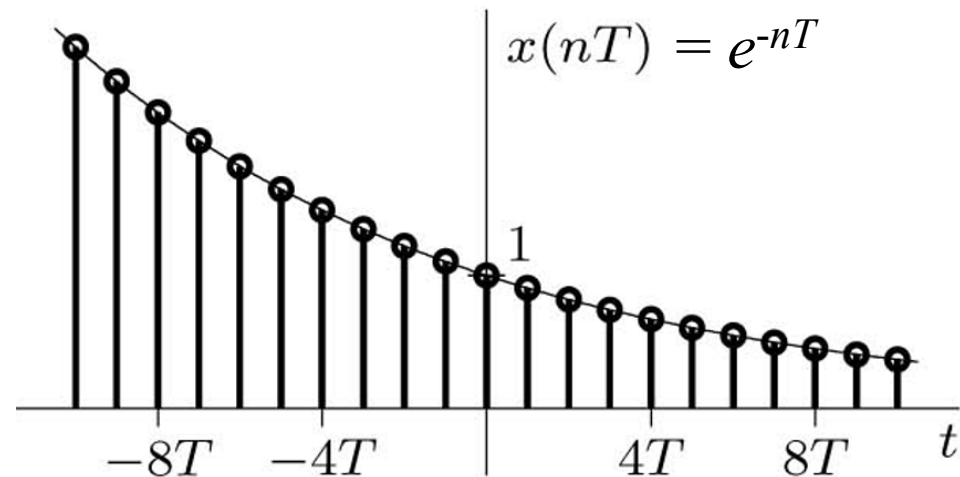
- Continuous to Discrete Signal
- DT Signal Models
- Signal Operations
- DT Signal Classifications
- DT Systems and Examples
- DT System Properties
- Digital Resampling

Exponential Continuous Signal to Discrete

$$x(t) = e^{-t}$$

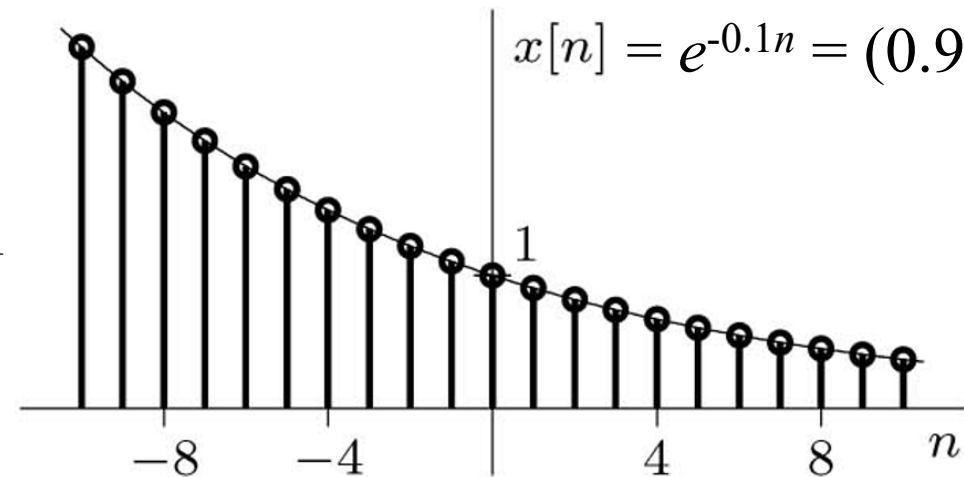


$$x(nT) = e^{-nT}$$



$$T = 0.1$$

$$x[n] = e^{-0.1n} = (0.905)^n$$



Sinusoidal Continuous Signal to Discrete

$$x(t) = \cos(\omega t) = \cos(2\pi f t)$$

$$f = 2 \text{ cycles/sec (Hz)}$$

$$f_s = 11 \text{ samples/sec}$$

$$T = 1/f_s = 1/11 \text{ second}$$

$$t \rightarrow Tn$$

$$x[n] = \cos(\omega T n)$$

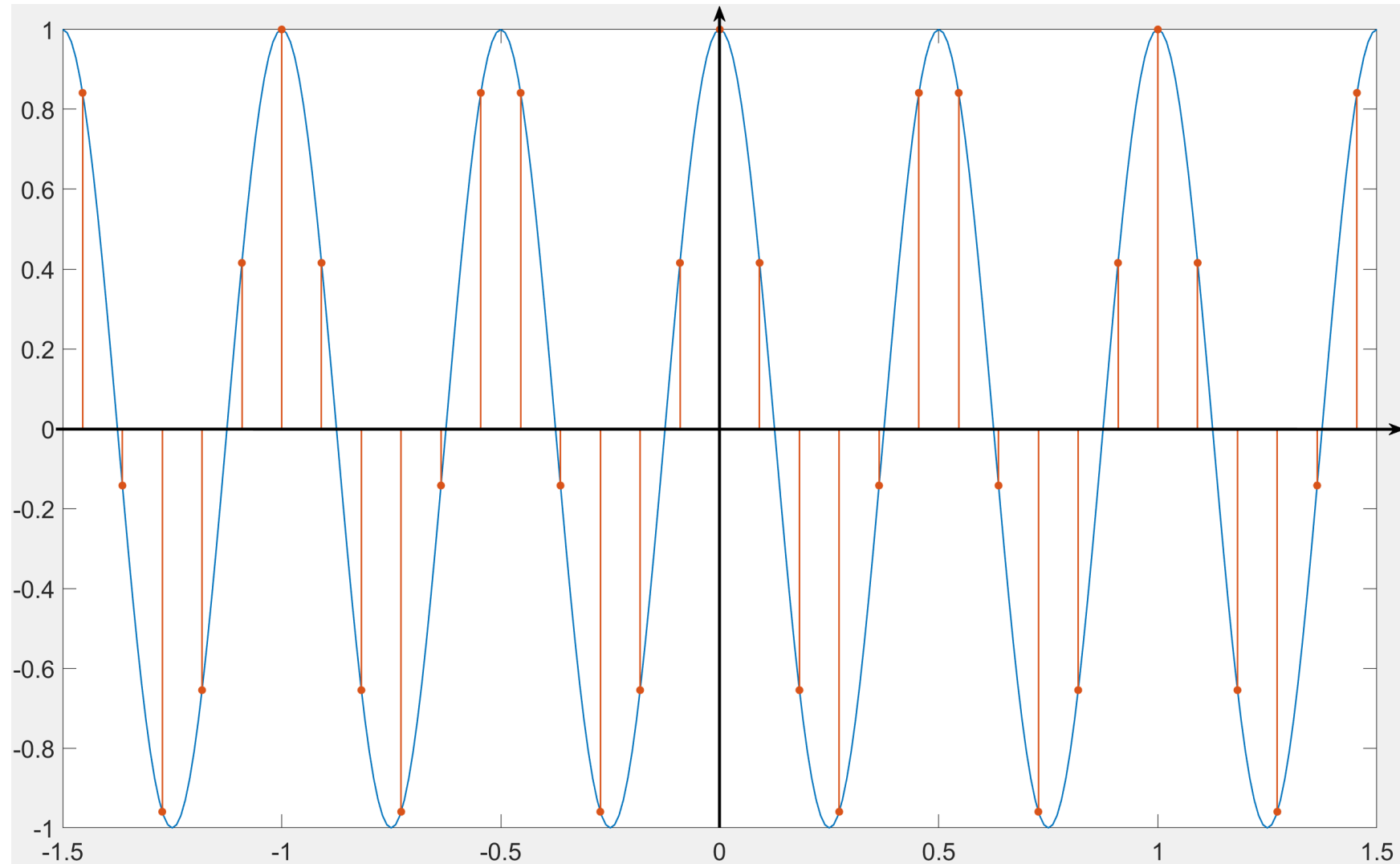
$$x[n] = \cos(\Omega n)$$

$$x[n] = \cos\left(2\pi \frac{f}{f_s} n\right)$$

$$x[n] = \cos(2\pi F n)$$

$$F = 2/11 \text{ cycles/samples}$$

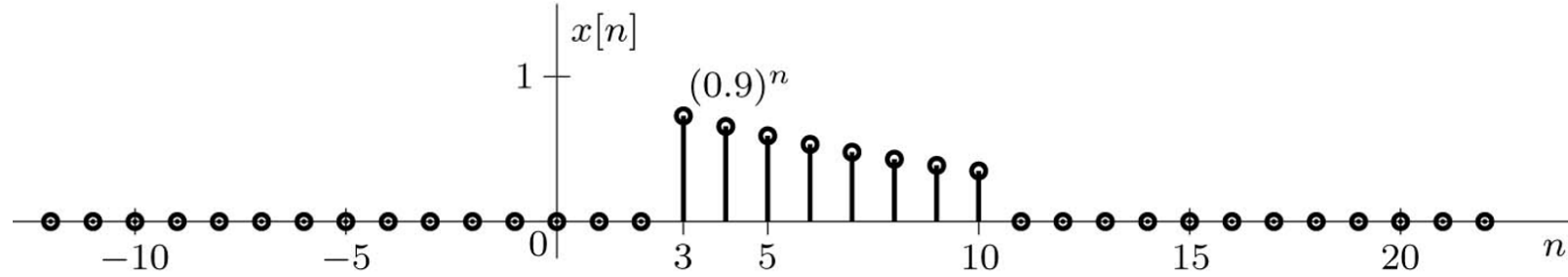
The unit of Ω is radians per samples



Operation on the Independent DT Variable

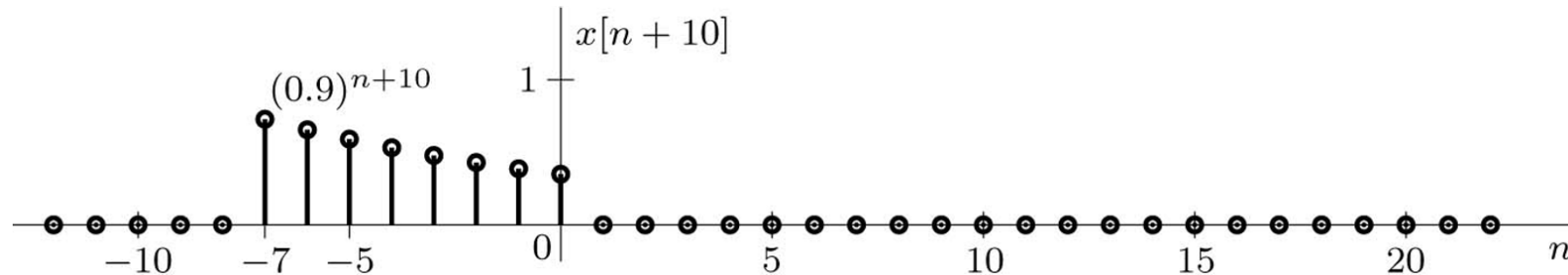
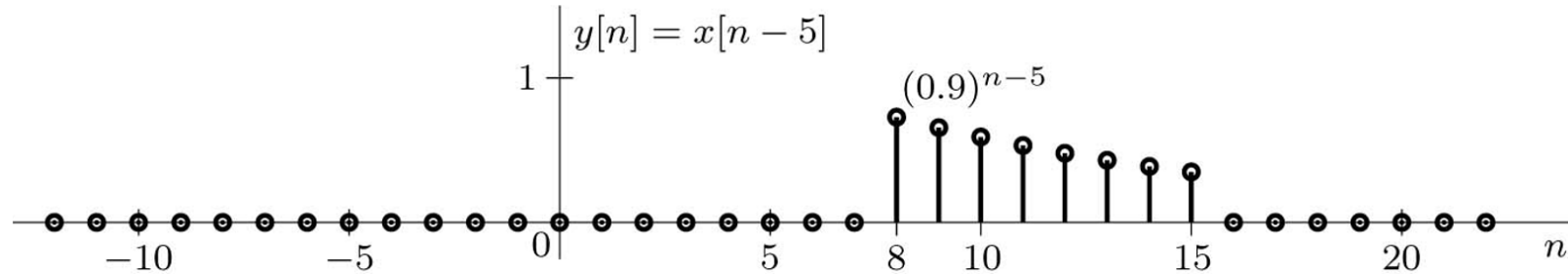
DT Time Shifting

$$x[n] = (0.9)^n \quad 3 \leq n \leq 10$$



$$y[n] = x[n - 5] = (0.9)^{n-5}$$

over $3 < n - 5 < 10$
or $8 \leq n \leq 15$



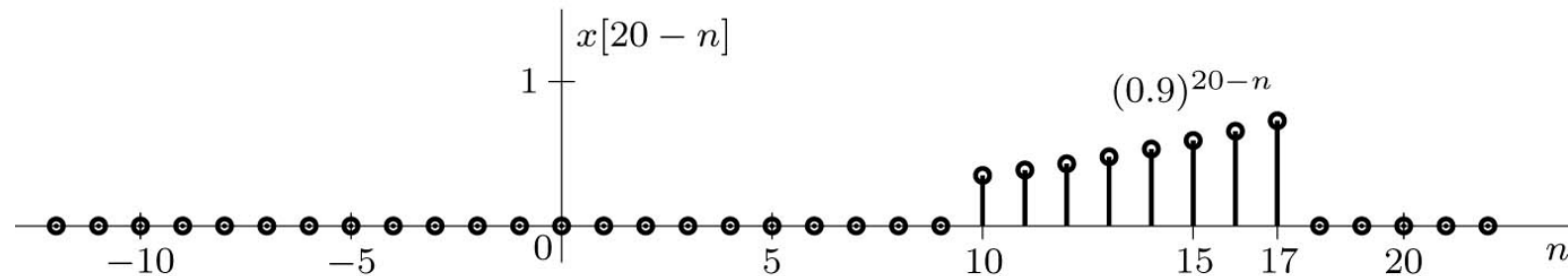
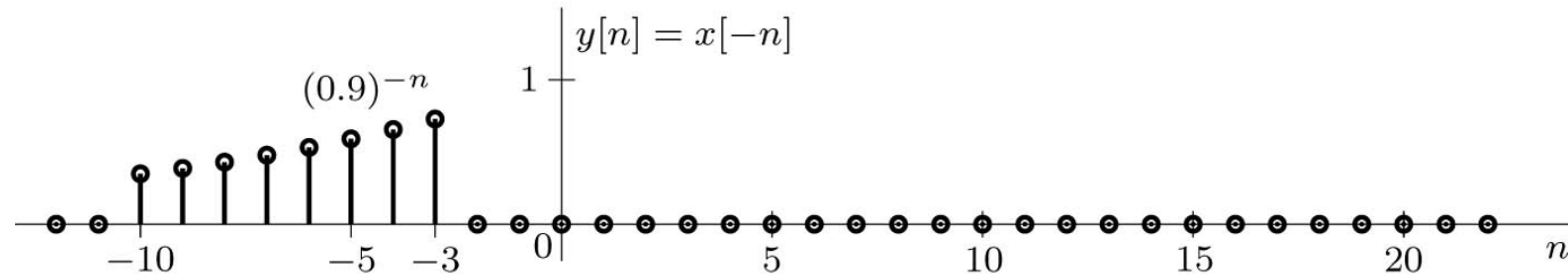
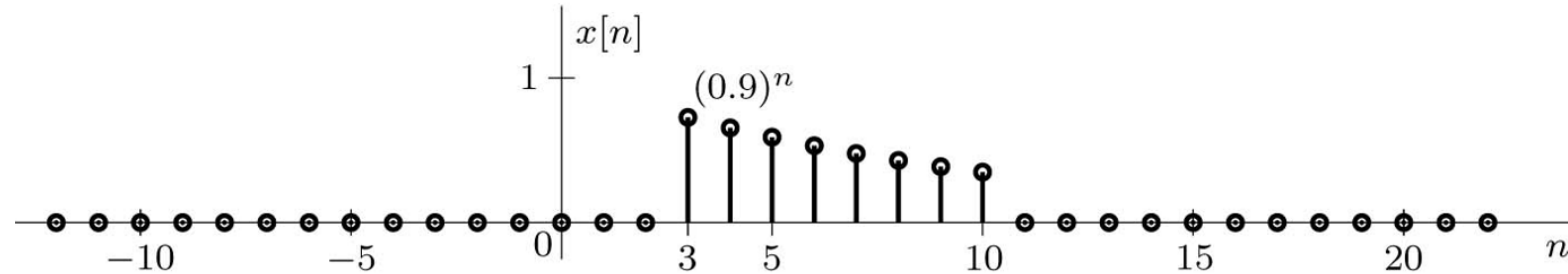
Operation on the Independent DT Variable

DT Time Reversal

Replace n by $-n$

Example: Plot $y[n] = x[20 - n]$

$$y[n] = x[-(n - 20)]$$



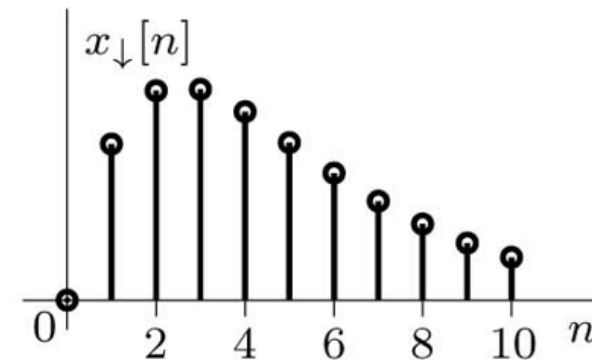
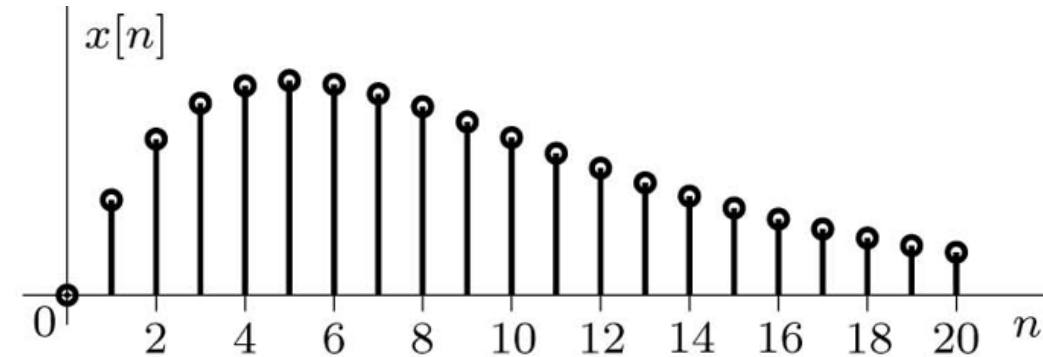
DT Time Scaling: Sampling Rate Conversion

Compression: Downsampling and Decimation

Replacing n with Mn in $x[n]$ compresses the signal by factor M to produce

$$x_{\downarrow}[n] = x[Mn].$$

This means that $x[Mn]$ selects every M th sample of $x[n]$ and deletes all the samples in between.



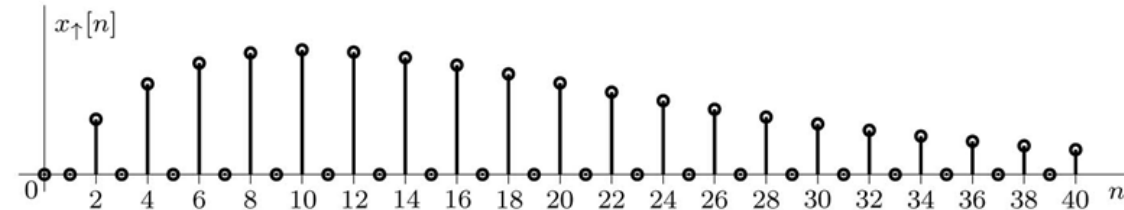
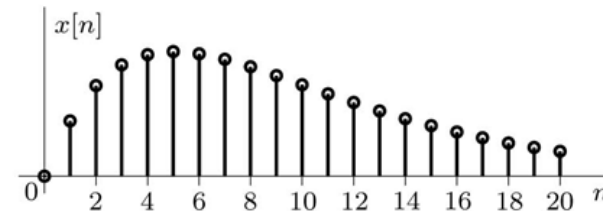
DT Time Scaling: Sampling Rate Conversion

Expansion: Upsampling, and Interpolation

An *interpolated* signal is generated in two steps: an expansion followed by filtering. Expand $x[n]$ by an integer factor L to obtain the expanded signal $x_{\uparrow}[n]$

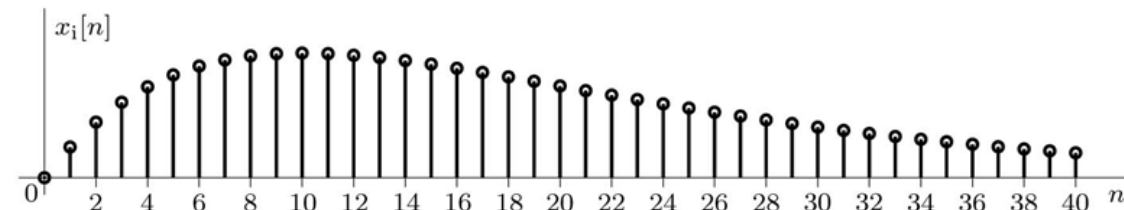
$$x_{\uparrow}[n] = \begin{cases} x[n/L] & n = 0, \pm L, \pm 2L, \dots \\ 0 & \text{otherwise} \end{cases}$$

$$x[0], \underbrace{0, 0, \dots, 0, 0}_{L-1 \text{ ZEROS}}, x[1], \underbrace{0, 0, \dots, 0, 0}_{L-1 \text{ ZEROS}}, x[2], \underbrace{0, 0, \dots, 0, 0}_{L-1 \text{ ZEROS}}, \dots$$



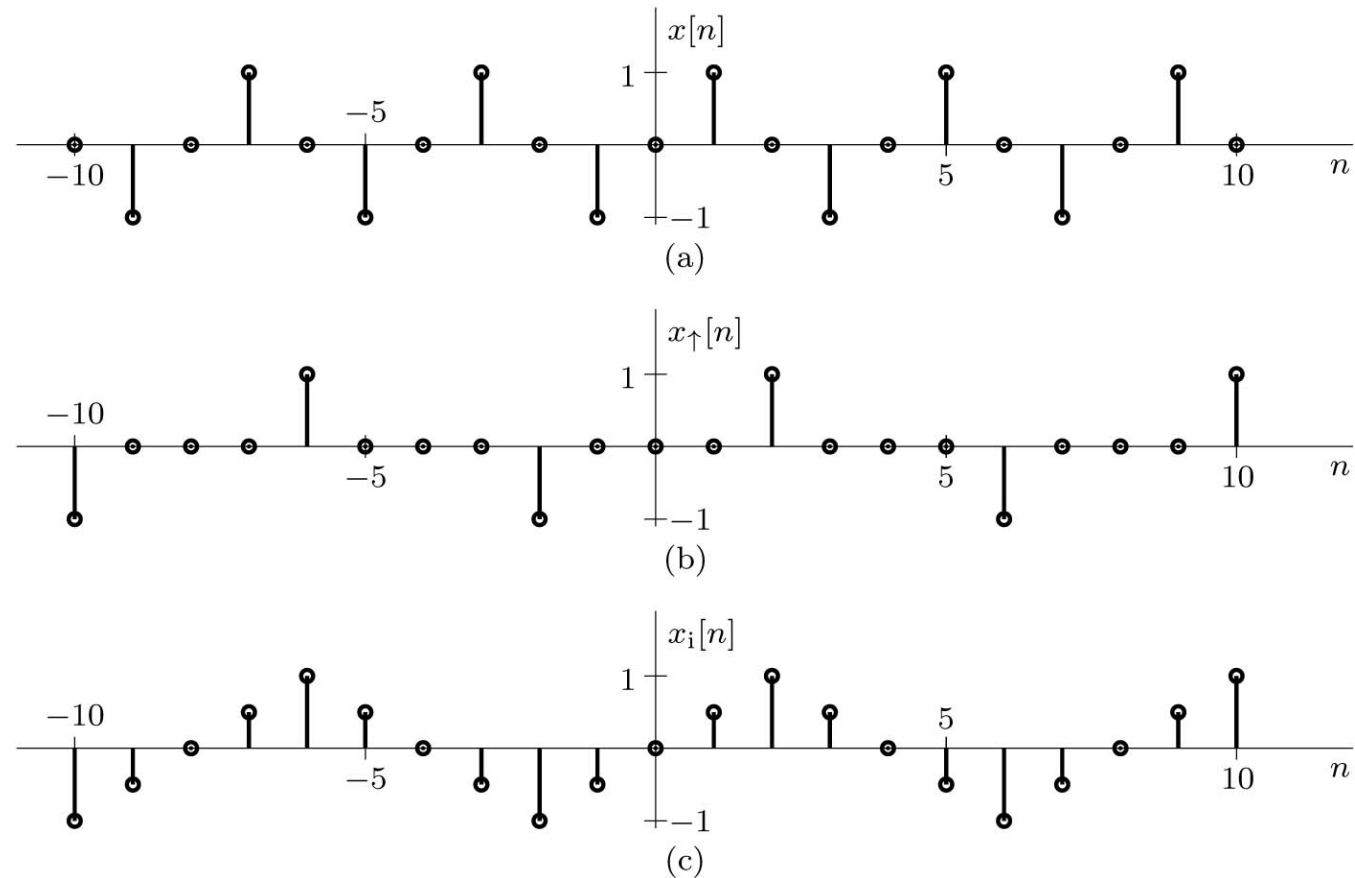
Linear Interpolation

$$y_i[n] = y_{\uparrow}[n] + (y_{\uparrow}[n-1] + y_{\uparrow}[n+1])/2$$



Example: Upsampling and Interpolation

A signal $x[n] = \sin(2\pi n/4)$ is expanded by $L = 2$ to produce $x_{\uparrow}[n]$. Next, this signal is interpolated according to $x_i[n] = 0.5x_{\uparrow}[n - 1] + x_{\uparrow}[n] + 0.5x_{\uparrow}[n + 1]$. Sketch $x[n]$, $x_{\uparrow}[n]$, and $x_i[n]$, and comment on the overall quality of the signal interpolation.

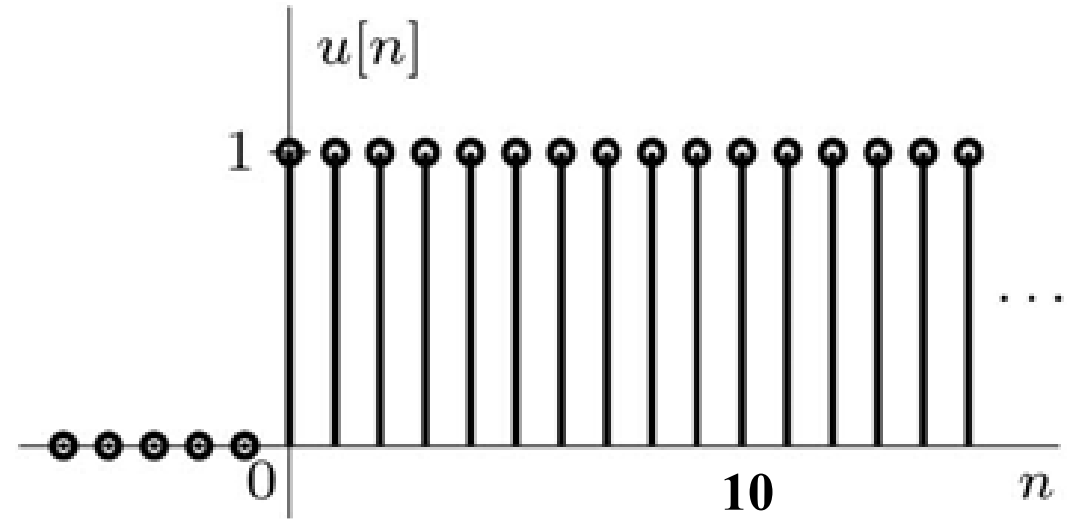


4.2

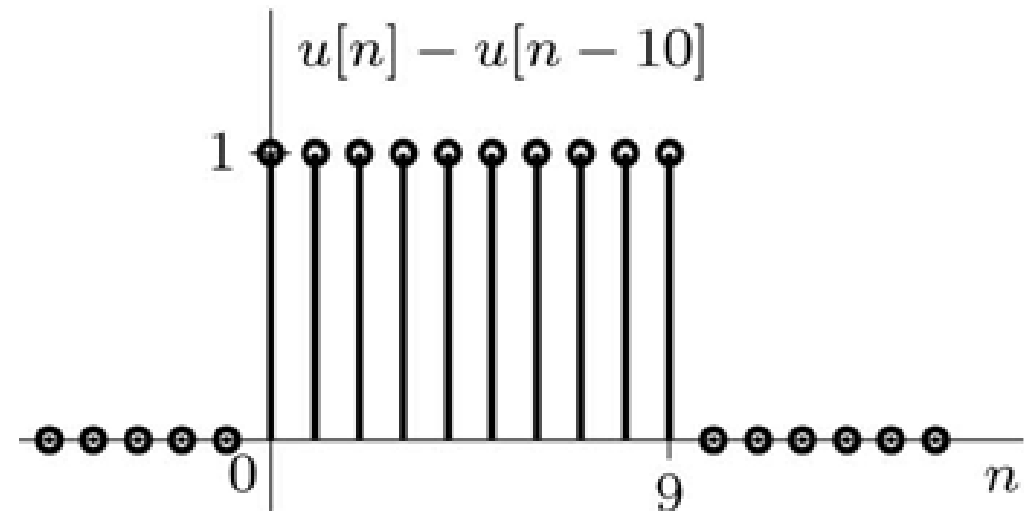
Signal Models

DT Signal Models: Unit Step Function $u[n]$

$$u[n] = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$$



Example: Plot $u[n] - u[n-10]$



DT Unit Impulse Function $\delta[n]$

$$\delta[n] = \begin{cases} 1 & n = 0 \\ 0 & n \neq 0 \end{cases}$$

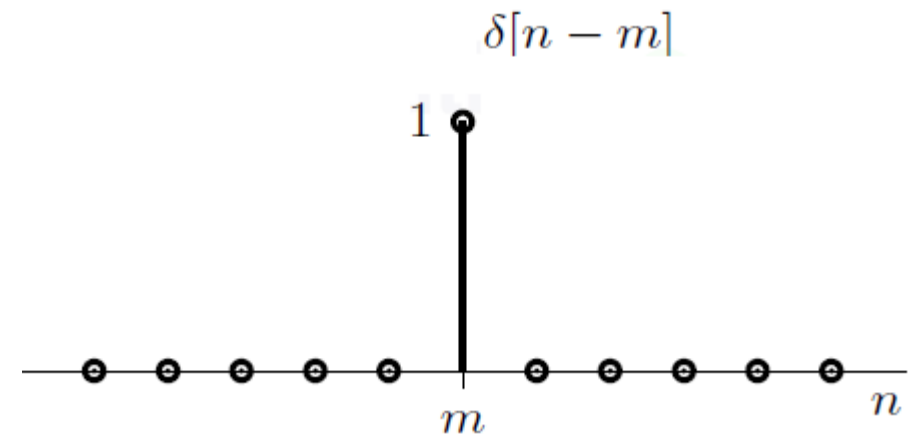
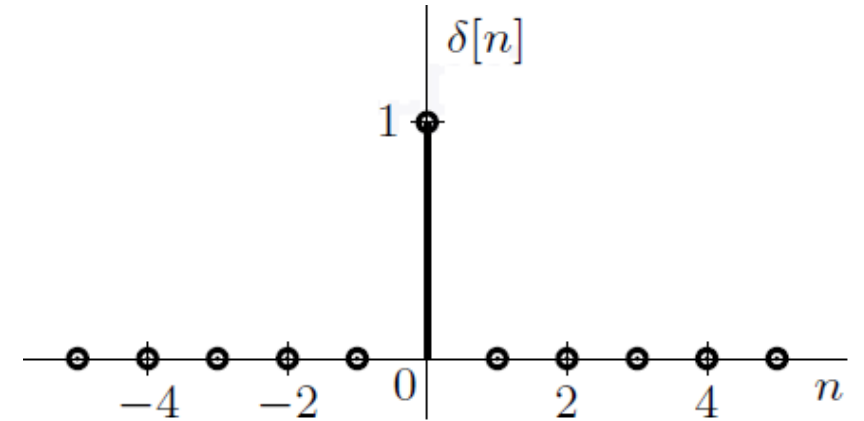
Multiplication by a DT Impulse $\delta[n]$

$$x[n]\delta[n - m] = x[m]\delta[n - m]$$

Relationship between $\delta[n]$ and $u[n]$

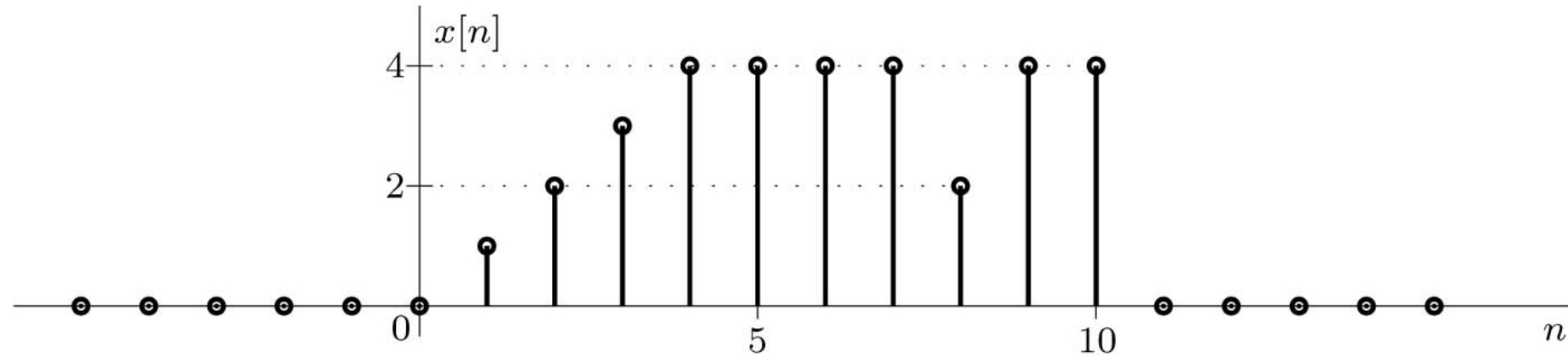
$$\text{Integral of } \delta[n] \quad u[n] = \sum_{m=-\infty}^n \delta[m] = \sum_{m=0}^{\infty} \delta[n - m]$$

$$\text{Derivative of } u[n] \quad \delta[n] = u[n] - u[n - 1]$$



Example

Describe the signal $x[n]$ shown below by a single expression, valid for all n , comprised of unit impulse and step functions



DT Sinusoid

The frequency of a DT sinusoid $\cos(\Omega n)$ must meet the following constraints: $\Omega \leq \pi$

$$\cos(\omega t) \rightarrow \cos(\omega T n) \rightarrow \cos(2\pi f/f_s n) \rightarrow \cos(2\pi F n) \rightarrow \cos(\Omega n) \quad \Omega = 2\pi f/f_s \leq \pi$$

The unit of Ω is **radians per samples**

The unit of F is **cycles per samples**

If we sample below the Nyquist rate, then $\Omega > \pi$ and the apparent frequency will be

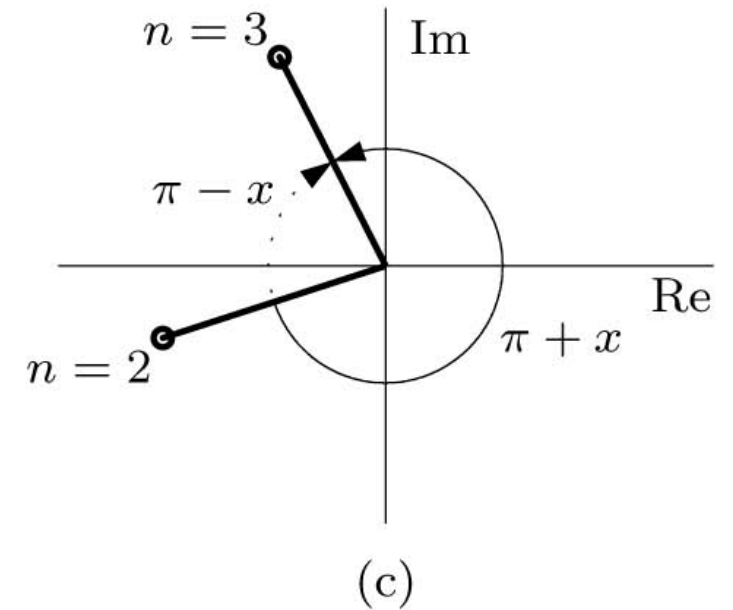
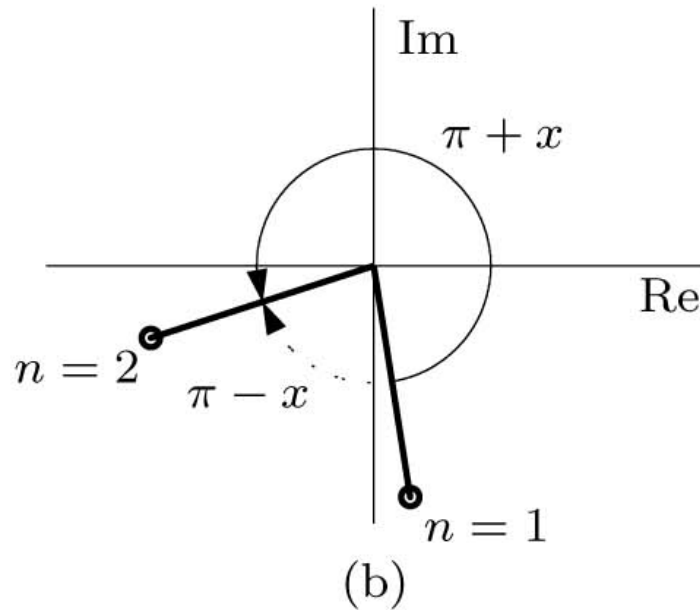
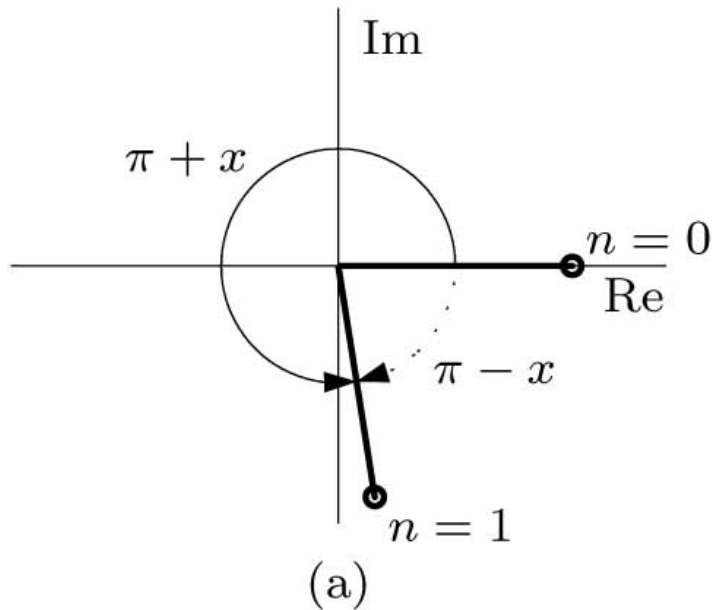
$$\Omega_a = |\langle \Omega + \pi \rangle_{2\pi} - \pi| \quad \text{Similar to} \quad f_a = |\langle f_o + f_s/2 \rangle_{f_s} - f_s/2|$$

Example: Express the following signals in terms of (positive) apparent frequency $|\Omega_a|$:

(a) $\cos(0.5\pi n + \theta)$ **(b)** $\cos(1.6\pi n + \theta)$ **(c)** $\cos(2.3\pi n + \theta)$

The Apparent Laziness of DT Exponential

Our Eyes acts like a lowpass filter and we will not see the fast motion in the counterclockwise direction. We will see slower motion in the clockwise direction. The actual frequency is $\pi + x$ but it will appear as $\pi - x$. (**wagon wheel effect**)



Progression of $e^{j(\pi+x)n}$ for n from (a) $0 \rightarrow 1$, (b) $1 \rightarrow 2$, and (c) $2 \rightarrow 3$

DT Exponential Function z^n

$$x(t) = e^{st} \xrightarrow[t \rightarrow Tn]{\text{Sampling}} x[n] = e^{sTn} = (e^{sT})^n = z^n$$

In general, s is a complex $s = \sigma + j\omega$, so $z = e^{\sigma T} e^{j\omega T}$

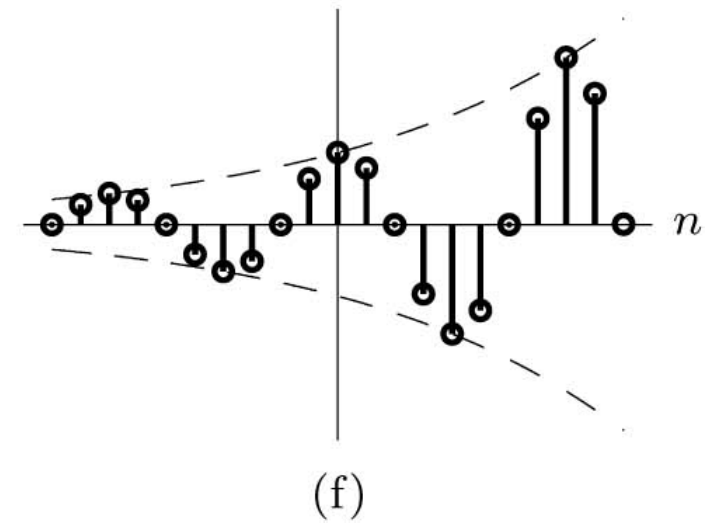
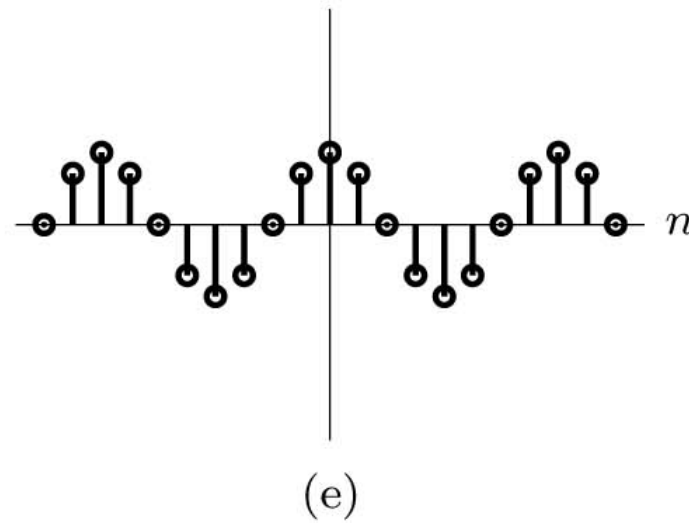
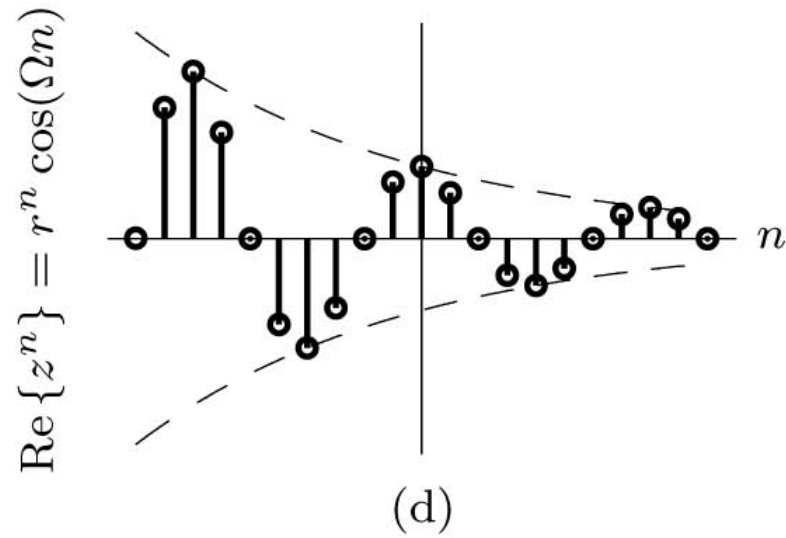
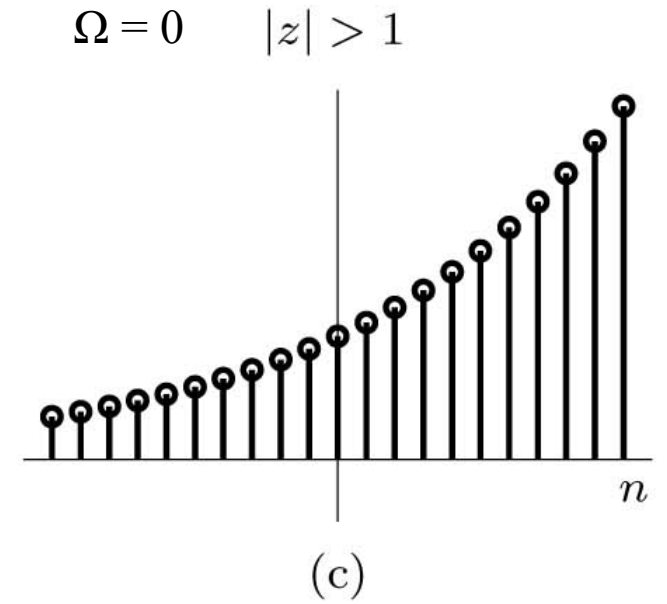
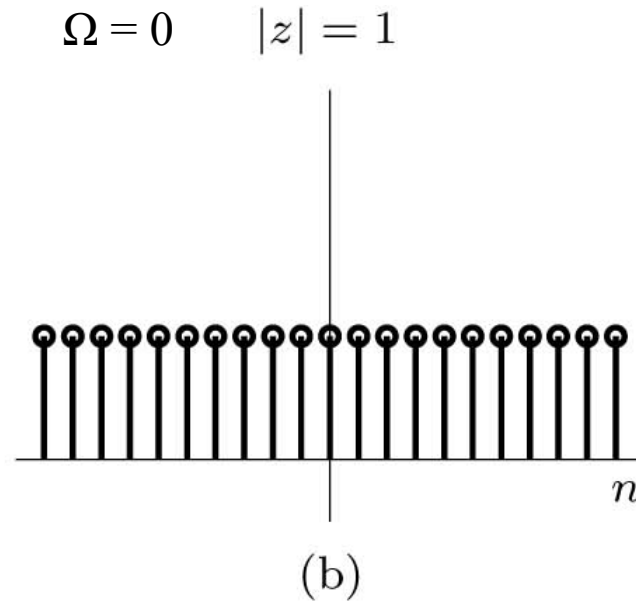
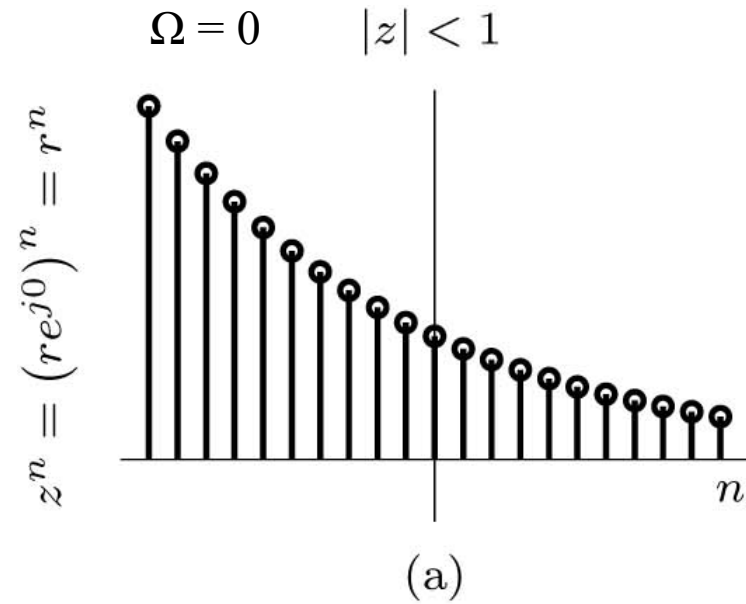
Letting $r = |z| = e^{\sigma T}$ and $\Omega = \angle z = \omega T$ so that $z = re^{j\Omega}$.

$$x[n] = (re^{j\Omega})^n = r^n e^{j\Omega n}$$

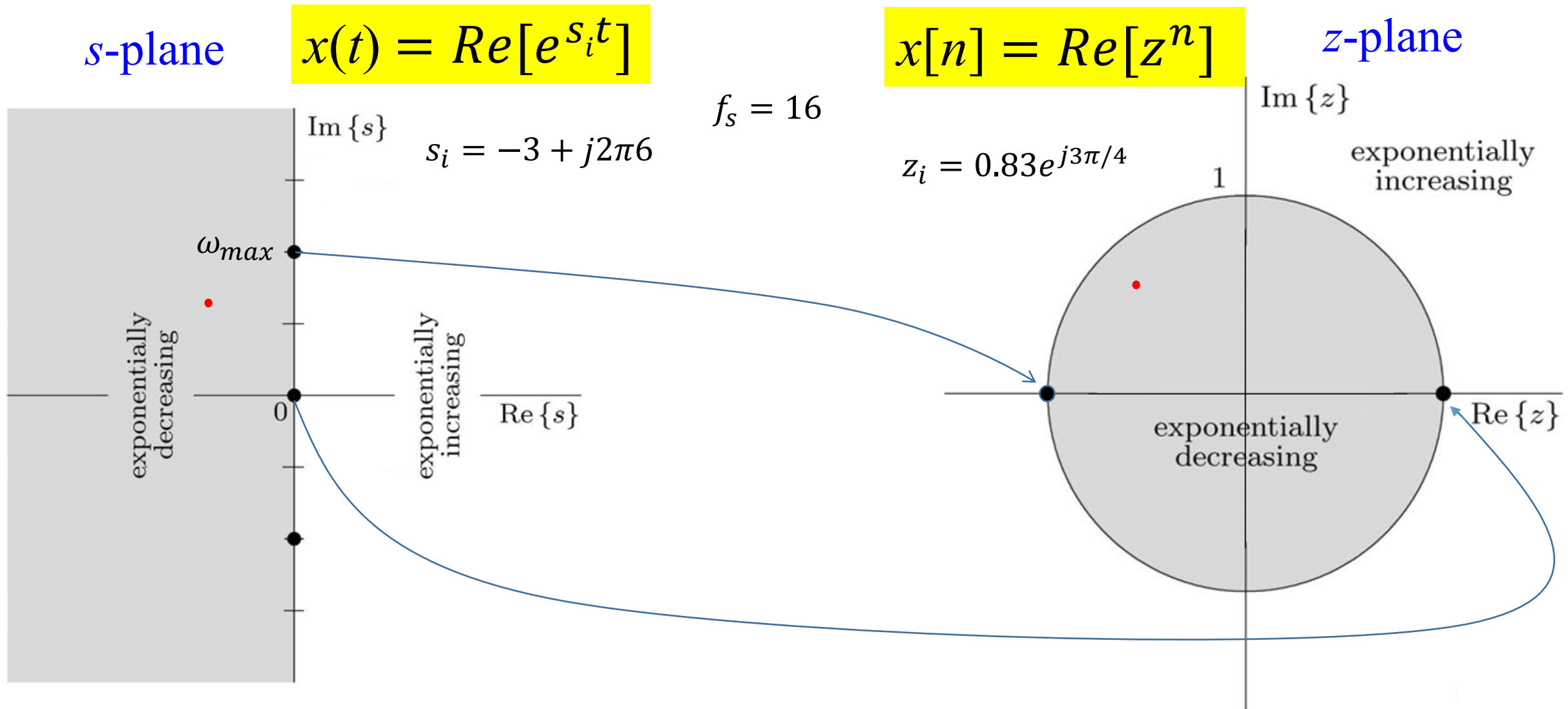
The following functions are special cases of z^n

1. a constant $k = k1^n$ (where $z = 1e^{j0}$),
2. a monotonic exponential r^n (where $z = re^{j0}$),
3. a sinusoid $\cos(\Omega n) = \text{Re} \{e^{j\Omega n}\}$ (where $z = 1e^{j\Omega}$), and
4. an exponentially varying sinusoid $r^n \cos(\Omega n) = \text{Re} \{r^n e^{j\Omega n}\}$ (where $z = re^{j\Omega}$)

DT Exponential Function $z^n = (re^{j\Omega})^n = r^n e^{j\Omega n}$



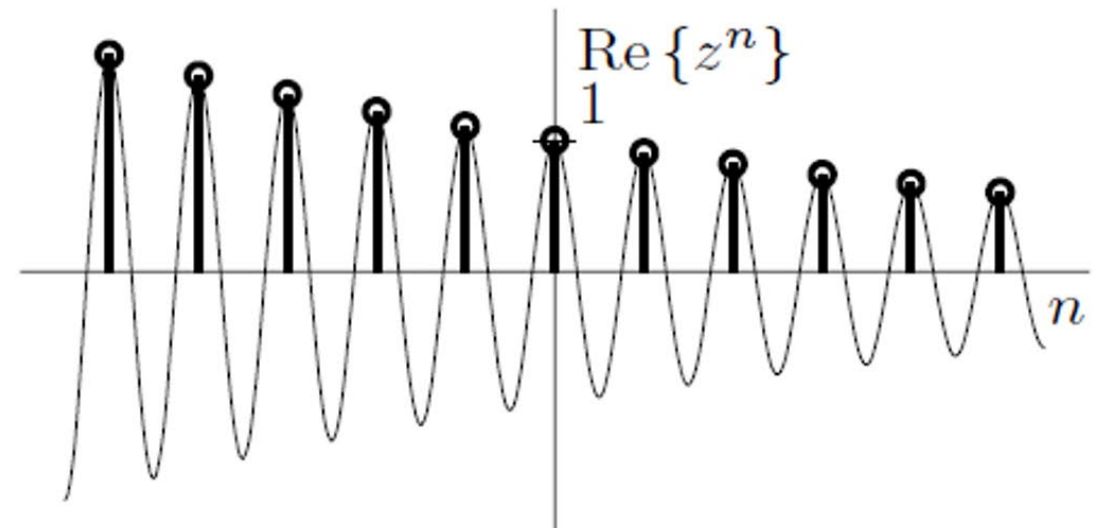
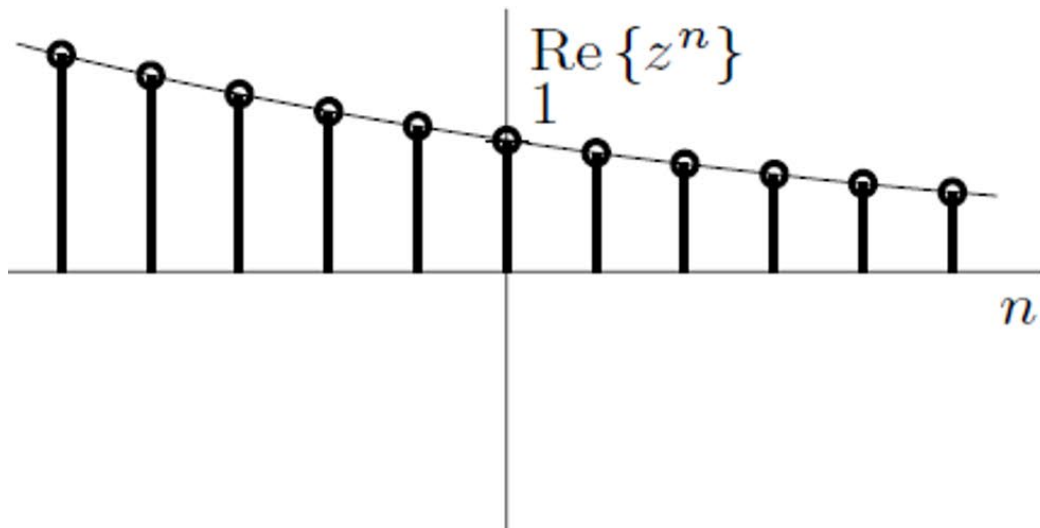
DT Exponential Function z^n in the Complex Plane



The s -plane imaginary axis becomes the z -plane *unit circle* $|z| = 1$.

Example

Consider the DT exponential $\text{Re} \{z^n\}$ obtained from sampling the signal $\text{Re} \{e^{st}\}$ with $T = 1$. Using $z = e^{sT}$, plot $\text{Re} \{z^n\}$, as well as $\text{Re} \{e^{st}\}$ for reference, over $-5 \leq n \leq 5$ for the cases $s = -0.1$ and $s = -0.1 + j2\pi$. Comment on the results.



$B = 1\text{Hz}$ so f_s should be 2 and $T = 0.5$ sec.

Note: Many points in the s-plane maps to one point in the z-plane ([aliasing](#)).

Homework 3 is due:

4.3

DT Signal Classification

1. Causal, Noncausal, and Anti-causal signals
2. Even and Odd Signals
3. Periodic and Aperiodic signals
4. Energy and Power Signals.

DT Signal Classifications

Causal, noncausal, and anti-causal signals,

- Causal is a signal that has values only at positive values of n

$$x[n] = 0 \text{ for } n < 0$$

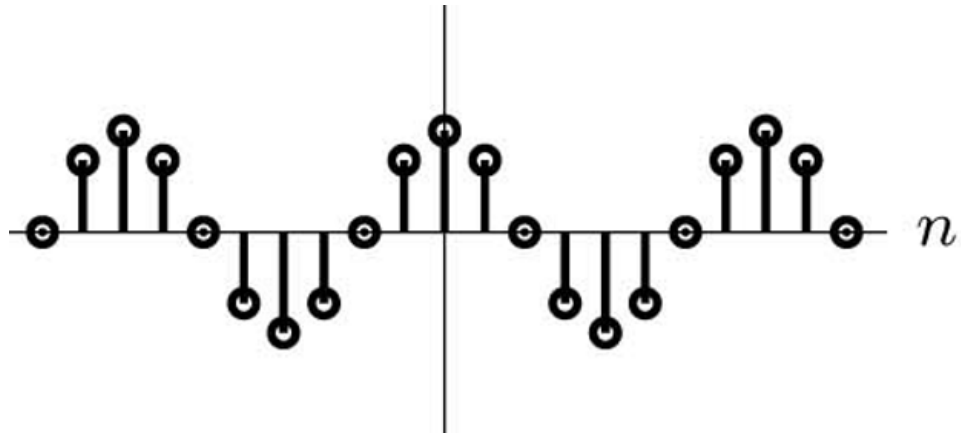
- Noncausal is a signal that has values at positive and negative values of n
- Anti-causal is a signal that has values only at negative values of n

$$x[n] = 0 \text{ for } n \geq 0$$

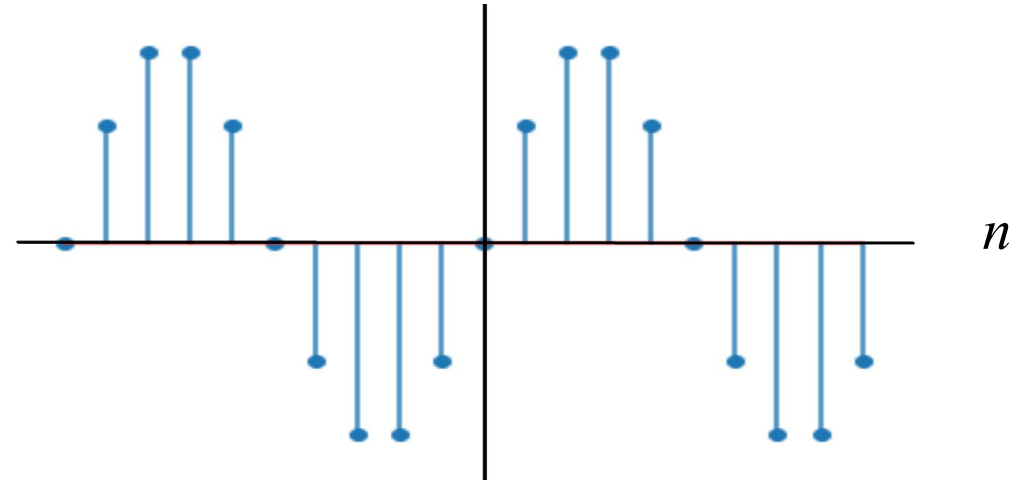
DT Signal Classifications

Even and Odd DT Signals

Even $x[n] = x[-n]$



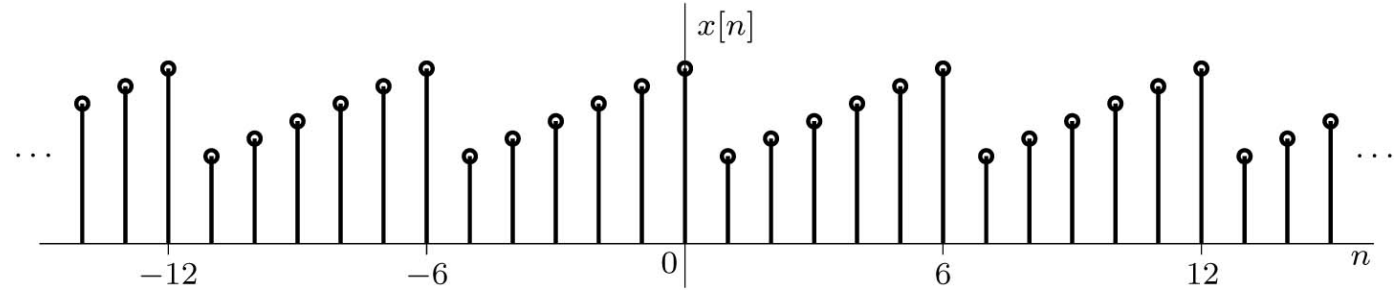
Odd $x[n] = -x[-n]$



DT Signal Classifications

Periodic and Aperiodic DT Signals

N -periodic $x[n] = x[n - N]$ for all n .



The smallest value of N that satisfy periodicity is called the fundamental period N_0 .

The fundamental frequency $F_0 = 1 / N_0$ or $\Omega_0 = 2\pi / N_0$

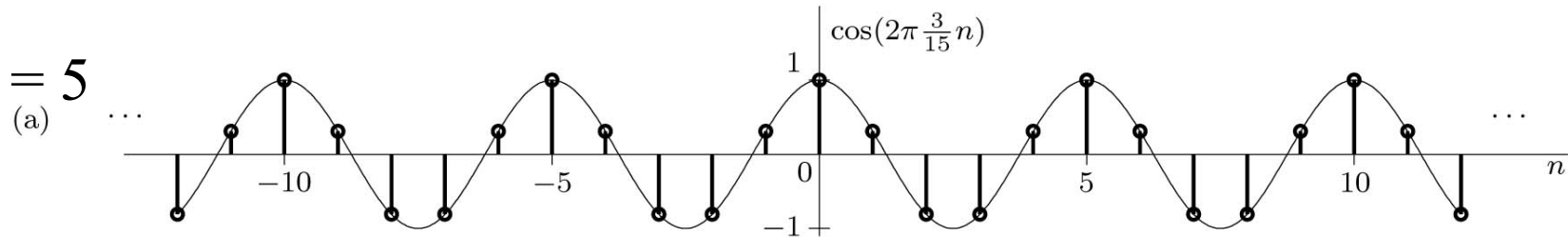
How do we find the fundamental period N_0 ?

The $\cos(2\pi F n)$ is discretely periodic if F can be expressed as $F = m / N_0$ where m is an integer and N_0 is the fundamental period. The ratio should be reduced, where the numerator and denominator are *coprime*.

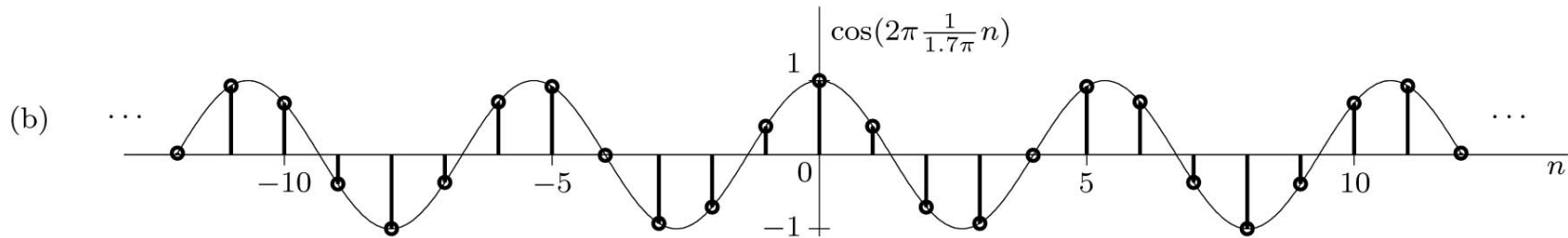
Example

Sketch the DT sinusoid $\cos(2\pi\mathcal{F}n)$ for the three cases (a) $\mathcal{F} = \frac{3}{15}$, (b) $\mathcal{F} = \frac{1}{1.7\pi}$, and (c) $\mathcal{F} = \frac{1}{5.5}$. In each case, determine whether or not the signal is periodic. If it is, determine the fundamental period N_0 , and state whether the fundamental frequency equals the sinusoid's frequency $\mathcal{F}$.

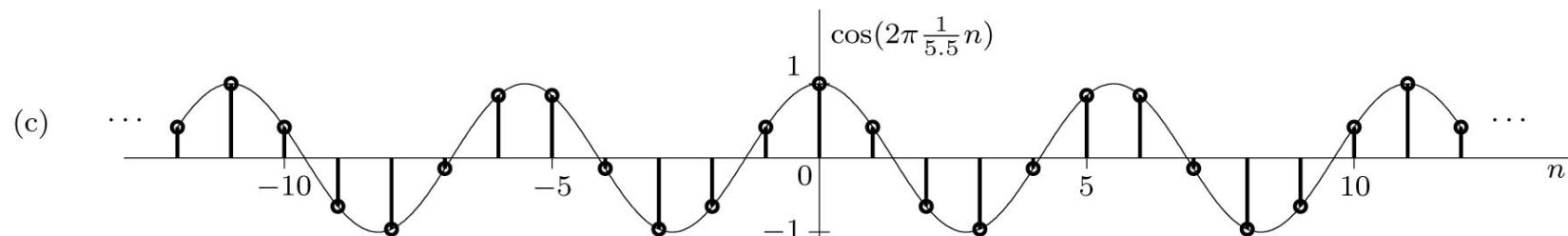
a) Periodic, $F = 3/15 = 1/5$, so $N_0 = 5$



b) Not periodic, F cannot be expressed as rational number.



c) Periodic, $F = 1/5.5 = 2/11$, so $N_0 = 11$



DT Signal Classifications

DT Energy Signals

$$E_x = \sum_{n=-\infty}^{\infty} |x[n]|^2 \quad x[n] \rightarrow 0 \text{ as } |n| \rightarrow \infty$$

DT Power Signal

$$P_x = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x[n]|^2$$

For periodic signal

$$P_x = \frac{1}{N_0} \sum_{n=0}^{N_0-1} |x[n]|^2$$

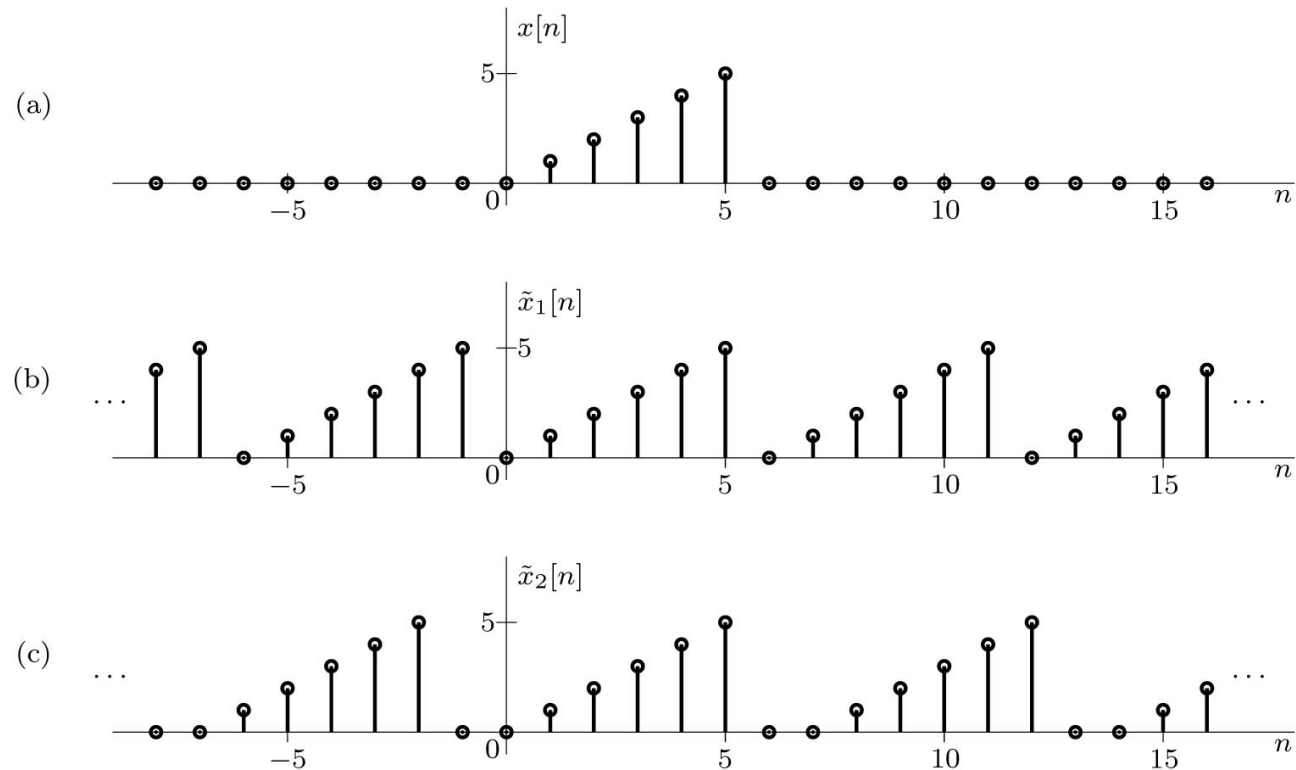
Example

Determine the energy and power for each of these three DS signals.

$$E_x = \sum_{n=0}^5 n^2 = 1^2 + 2^2 + 3^2 + 4^2 + 5^2 = 55$$

$$P_{\tilde{x}_1} = \frac{1}{6} \sum_{n=0}^5 n^2 = \frac{1}{6} E_x = \frac{55}{6} \approx 9.1667$$

$$P_{\tilde{x}_2} = \frac{1}{7} \sum_{n=0}^5 n^2 = \frac{1}{7} E_x = \frac{55}{7} \approx 7.8571$$



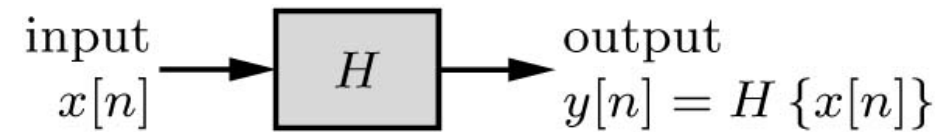
4.4

DT Systems and Examples

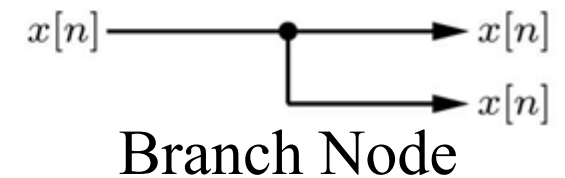
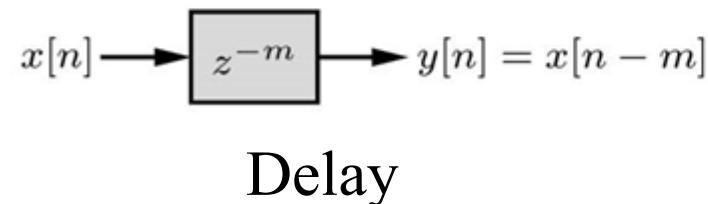
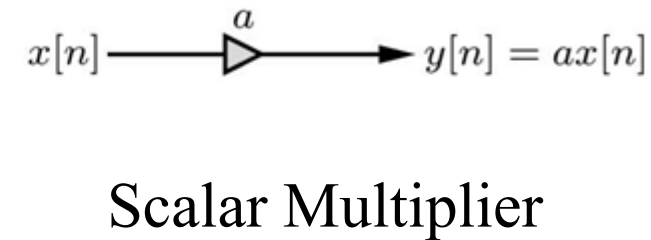
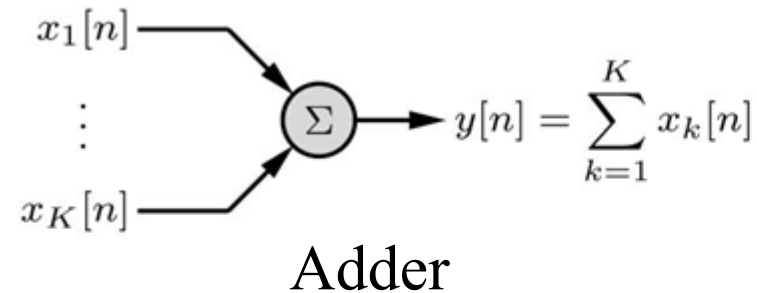
DT Systems and Examples

Discrete-time systems process discrete-time signals (inputs) to yield another set of discrete time signals (outputs). Operator H is used to represent the system function.

$$y[n] = H\{x[n]\}$$



The four major blocks in any DT systems are

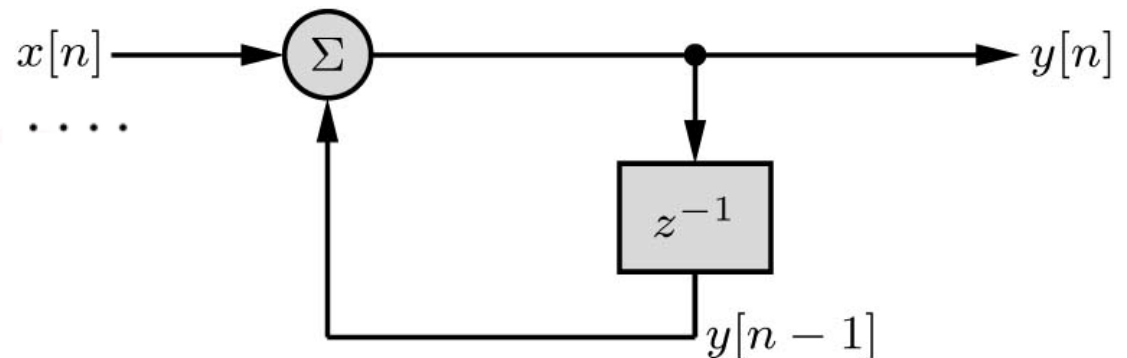


Recursive and Non-Recursive Systems

Non-recursive System: Output is calculated from current and previous inputs.

$$y[n] = \sum_{k=1}^n x[k] \quad n = 1, 2, 3, \dots$$

Recursive System: Output is calculated from previous output and current input.

$$y[n] = y[n-1] + x[n] \quad n = 1, 2, 3, \dots$$
$$= \underbrace{\sum_{k=1}^{n-1} x[k]}_{y[n-1]} + x[n]$$


DT Systems and Examples

Delay and Advance Forms

$$\text{Delay} \quad y[n] - y[n - 1] = x[n] \quad n = 0, 1, 2, \dots$$

$$\text{Advance} \quad y[n + 1] - y[n] = x[n + 1] \quad n = 0, 1, 2, \dots$$

Advance form is noncausal and unrealizable but can be used to predict future value if we are willing to wait for the output.

The Accumulator

$$y[n] = \sum_{k=-\infty}^n x[k] \quad y[n] - y[n - 1] = x[n]$$

Accumulator is analogous to the integrator in continuous-time system

Example (Bank Balance)

A person makes a deposit (the input) in a bank regularly at an interval of $T = 1$ month. The bank pays a certain interest on the account balance during the period T and mails out a periodic statement of the account balance (the output) to the depositor. Find the equation relating the output $y[n]$ (the balance) to the input $x[n]$ (the deposit). Assuming that a person invests \$1000 monthly starting at $n = 1$ and earns a 0.5% monthly interest rate, how much money is earned at 5 years ($n = 60$)? Provide a block diagram representation of the system.

Solution

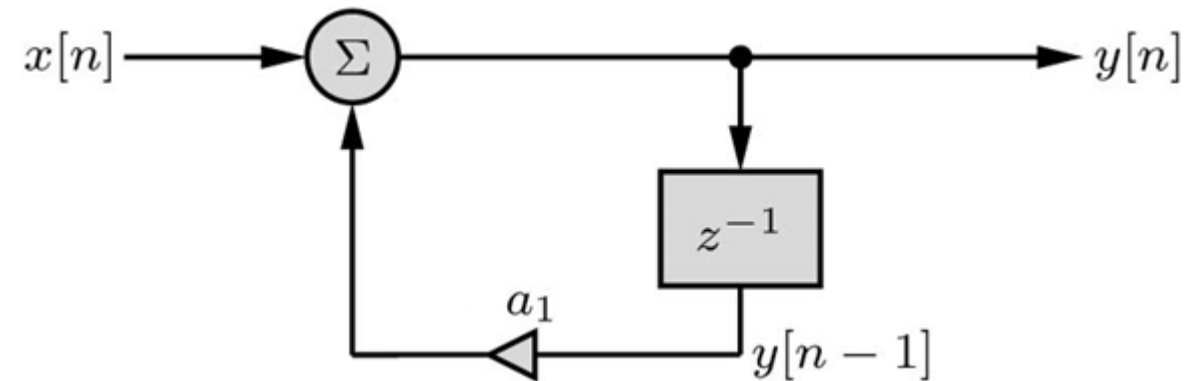
$x[n]$ = the n th deposit

$y[n]$ = the month- n account balance, including the n th deposit

r = interest rate per period T

$$\begin{aligned} y[n] &= y[n-1] + ry[n-1] + x[n] \\ &= (1+r)y[n-1] + x[n] \end{aligned}$$

Let $a_1 = (1+r)$ $y[n] = a_1 y[n-1] + x[n]$



Example (Digital Differentiator)

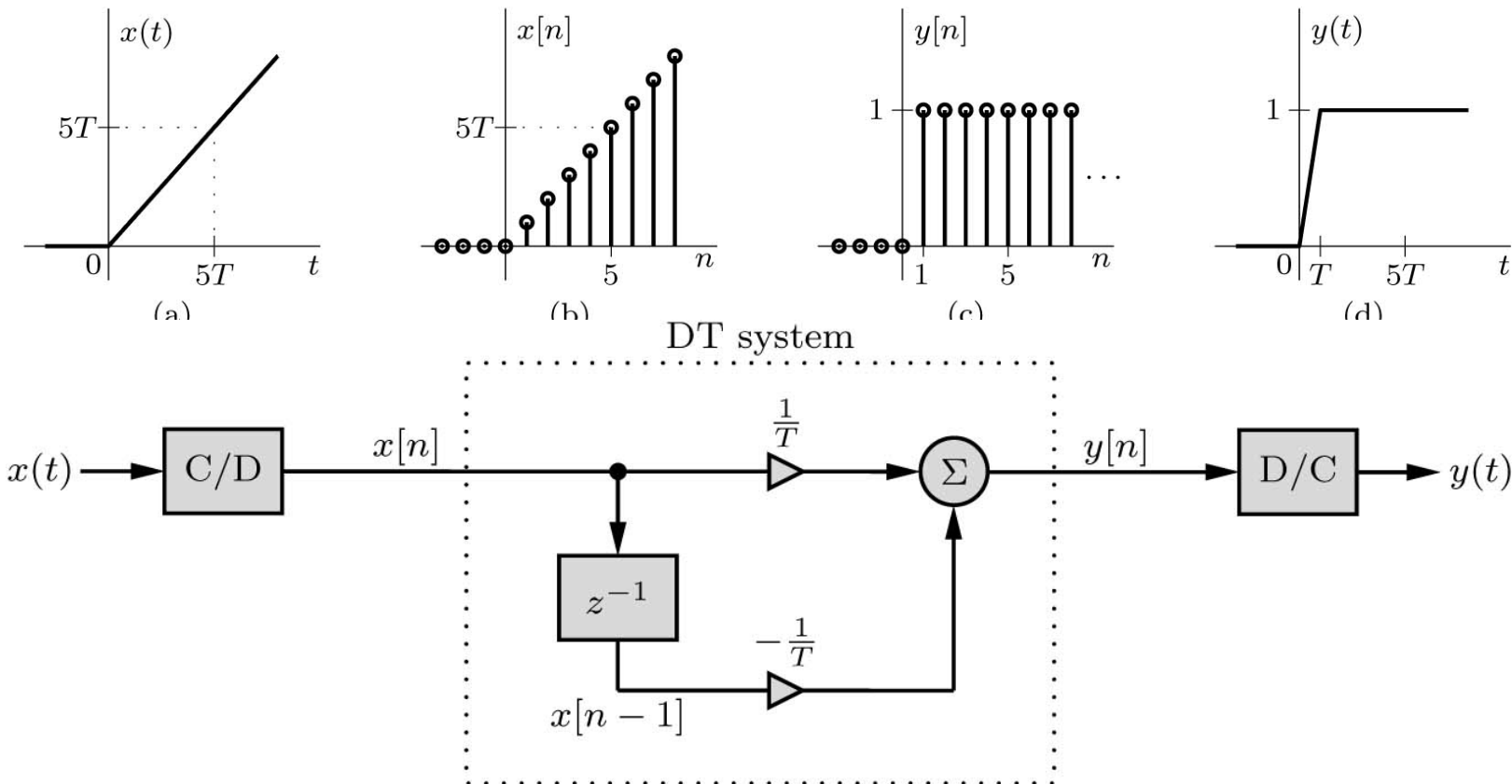
Design a discrete-time system to differentiate continuous-time signals. Construct a block diagram representation of the system and investigate its response to a ramp input $x(t) = tu(t)$. Suggest an appropriate sampling interval if this is used in an audio system where the input signal bandwidth is below 20 kHz.

Solution

$$y[n] = \lim_{T \rightarrow 0} \frac{x[n] - x[n-1]}{T}$$

$$y[n] = \frac{1}{T}x[n] - \frac{1}{T}x[n-1]$$

$$T \leq \frac{1}{2f_{\max}} = \frac{1}{40000} = 25 \mu\text{s}$$

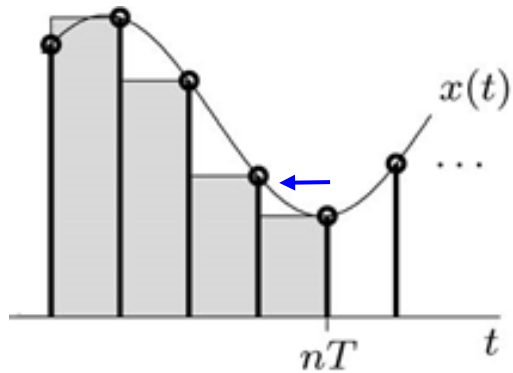


Example (Digital Integrator)

Design discrete-time systems to integrate continuous-time signals using **(a)** a backward-staircase approximation, **(b)** a forward-staircase approximation, and **(c)** a trapezoidal approximation. Construct a block diagram representation for each system.

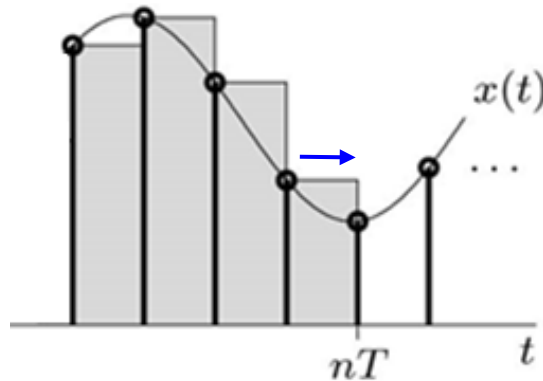
Solution

Backward



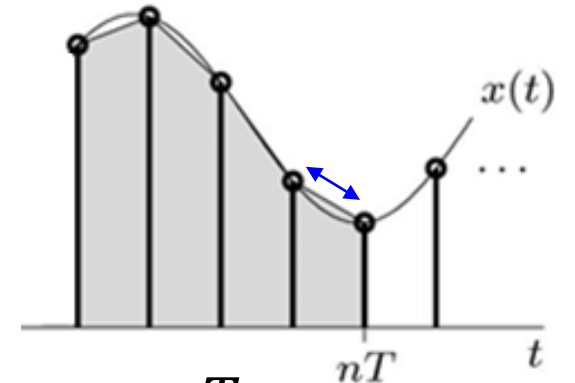
$$y[n] = y[n - 1] + Tx[n]$$

Forward



$$y[n] = y[n - 1] + Tx[n - 1]$$

Trapezoidal

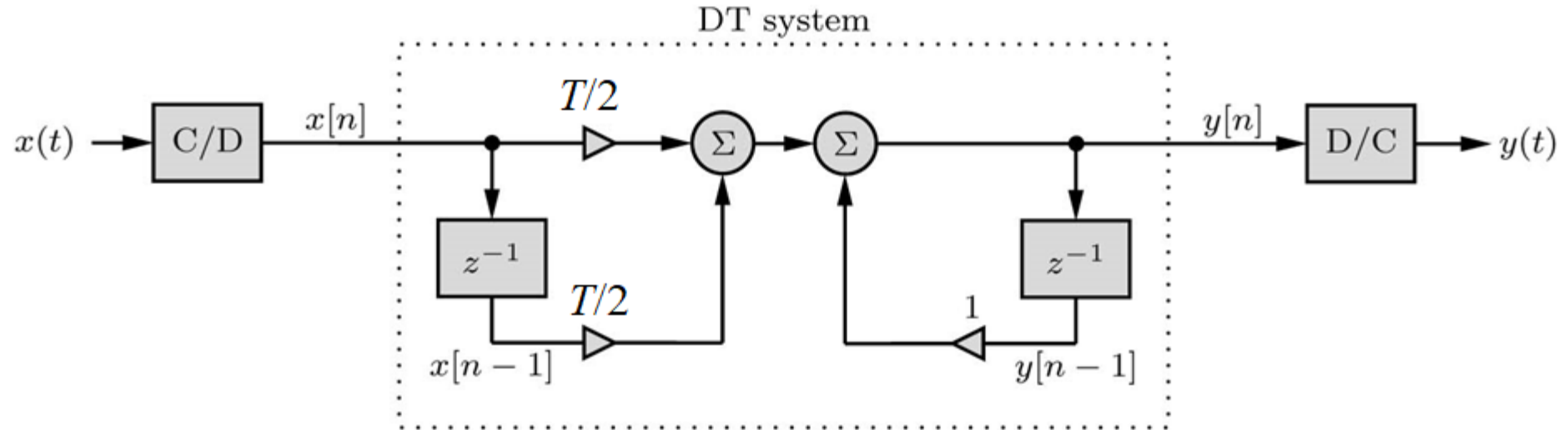


$$y[n] = y[n - 1] + \frac{T}{2} (x[n] + x[n - 1])$$

Example (Digital Integrator)

Trapezoidal

$$y[n] = y[n - 1] + \frac{T}{2} (x[n] + x[n - 1])$$



Order and General Form of Difference Equations

Just as the highest-order derivative of the input or output represents the order of a differential equation, the highest-order difference of the input or output represents the *order* of a difference equation. Order is reflective of system complexity. For CT systems, **order indicates the number of energy storage elements**, such as capacitors and inductors, required for implementation. For DT systems, **order indicates the required amount of memory**.

$$\frac{d}{dt}y(t) + cy(t) = x(t)$$

$$\lim_{T \rightarrow 0} \frac{y[n] - y[n-1]}{T} + cy[n] = x[n]$$

$$y[n] + \alpha y[n-1] = \beta x[n]$$

$$\alpha = \frac{-1}{1 + cT} \quad \text{and} \quad \beta = \frac{T}{1 + cT}$$

In advance form

$$y[n+1] + \alpha y[n] = \beta x[n+1]$$

$$\begin{aligned} \left. \frac{d^2}{dt^2} y(t) \right|_{t=nT} &= \lim_{T \rightarrow 0} \frac{1}{T} \left(\left. \frac{d}{dt} y(t) \right|_{t=nT} - \left. \frac{d}{dt} y(t) \right|_{t=(n-1)T} \right) \\ &\approx \frac{1}{T} \left(\frac{y[n] - y[n-1]}{T} - \frac{y[n-1] - y[n-2]}{T} \right) \\ &= \frac{1}{T^2} (y[n] - 2y[n-1] + y[n-2]). \end{aligned}$$

$$\left. \frac{d^K}{dt^K} y(t) \right|_{t=nT} \approx \frac{1}{T^K} \sum_{k=0}^K (-1)^k \binom{K}{k} y[n-k].$$

Order and General Form of Difference Equations

$$a_0 y[n + K] + a_1 y[n + K - 1] + \cdots + a_{K-1} y[n + 1] + a_K y[n] = \\ b_0 x[n + L] + b_1 x[n + L - 1] + \cdots + b_{L-1} x[n + 1] + b_L x[n]$$

Advance Form $\sum_{k=0}^K a_k y[n + K - k] = \sum_{l=0}^L b_l x[n + L - l]$ For stable system $K \geq L$

Example

Assuming a sampling interval $T = 0.1$, determine a difference equation model for the differential equation

$$\ddot{y}(t) + 4\dot{y}(t) + 3y(t) = 100x(t)$$

with initial conditions $y(0) = 0$ and $\dot{y}(0) = 10$.

Answer:

$$y[n] - \frac{240}{143}y[n - 1] + \frac{100}{143}y[n - 2] = \frac{100}{143}x[n]$$

with initial conditions $y[0] = 0$ and $y[-1] = -1$

Advantage of Digital Signal Processing

- Digital systems can tolerate considerable variation in signal values and, hence, are less sensitive to changes in the component parameter values.
- Digital systems are extremely flexible and easy to implement.
- Even in the presence of noise, reproduction with digital messages is extremely reliable, often without any deterioration whatsoever.
- Digital signals are easily and inexpensively stored electronically, without any loss of signal quality.
- Digital filters can be easily time shared and therefore can serve a number of inputs simultaneously.
- Digital implementation permits the use of a wide variety of hardware options, including computers and microprocessors, DSPs.

Disadvantage: Power consumption, limited frequency range and resolution, system complexity.

4.5

DT System Properties

DT System Properties

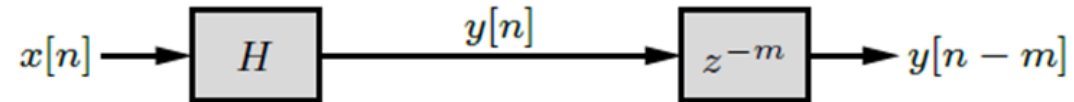
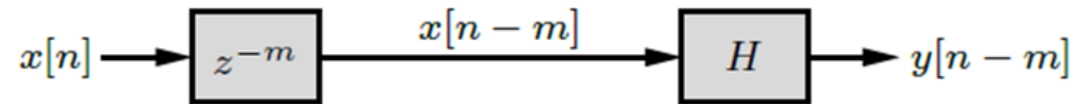
Linearity

If $c_1 x_1[n] \Rightarrow c_1 y_1[n]$ and $c_2 x_2[n] \Rightarrow c_2 y_2[n]$

then $c_1 x_1[n] + c_2 x_2[n] \Rightarrow c_1 y_1[n] + c_2 y_2[n]$

Time Invariance

$x[n] \Rightarrow y[n]$ implies that $x[n - m] \Rightarrow y[n - m]$



Example: A system is described by the equation $y[n] = e^{-n} x[n]$. Is the system time invariant?

DT System Properties

Causality:

A causal system is one for which the output at any instant $n = n_0$ depends only on the value of the input $x[n]$ for $n \leq n_0$.

$$h[n] = 0 \quad \text{for } n < 0$$

The general-order difference equation for stable system is

$$\sum_{k=0}^K a_k y[n - k] = \sum_{l=0}^L b_l x[n - l]$$

For a causal system $K \geq L$.

DT System Properties

Stability

External Stability (BIBO)

$$|y[n]| \leq K_y < \infty \quad \text{if} \quad |x| \leq K_x < \infty$$

Internal Stability (asymptotic and marginal stability): depends on the parameter of the system (Chapter 5)

Memory

If the output of a discrete-time system depends on input values other than the present one, then the system has *memory*. Systems with memory are also called *dynamic systems*. A system without memory is called a *memoryless, instantaneous, or static system*.

Invertibility

A discrete-time system H is invertible if an inverse system H_i exists such that the cascade of H and H_i results in an identity system (output equals input).

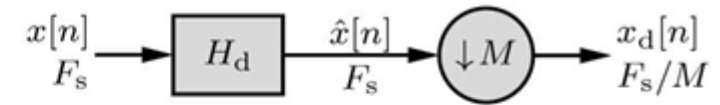
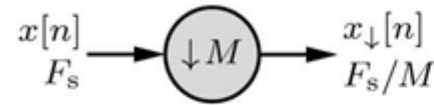
4.6

Digital Resampling

Digital Resampling

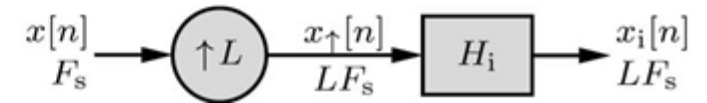
Combined with appropriate filtering, expansion and compression are the backbone operations for multirate systems. These operations provide an effective means to *digitally* alter the sampling rate of a DT signal, even by non-integer factors.

Compressor (Downsampling)



Antialiasing
Filter

Expander (Upsampling)



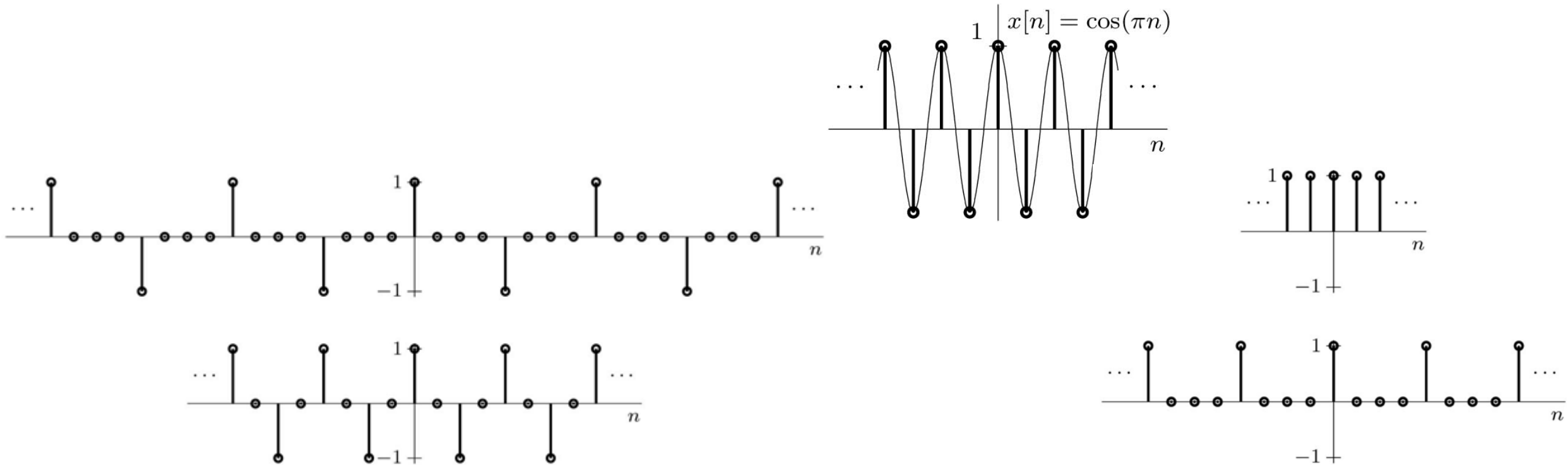
Interpolator
Filter

Cascade ordering of compression and expansion systems does matter in certain cases, see next example.



Example: Consider the signal $x[n] = \cos(\pi n)$, whose frequency is the highest possible without aliasing. Determine the output when this signal is applied to the following systems:

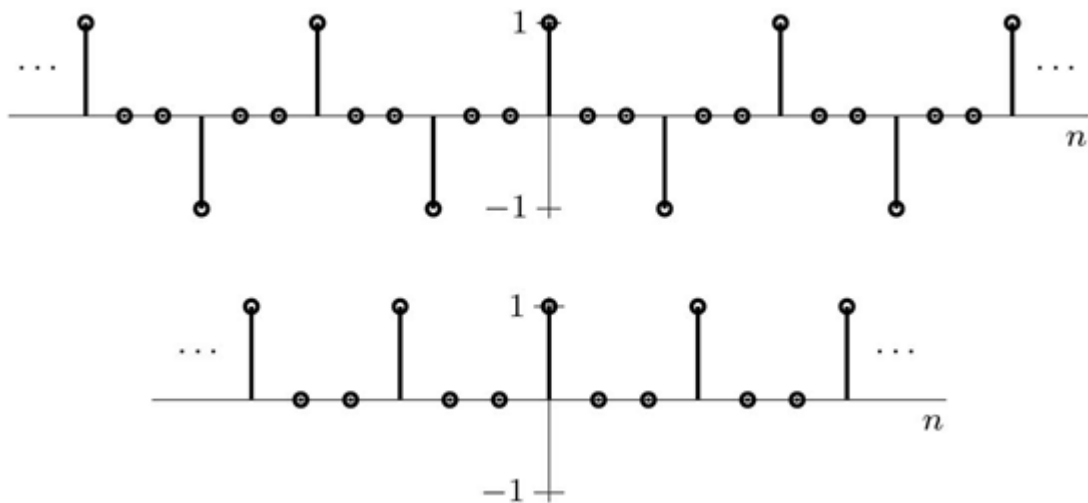
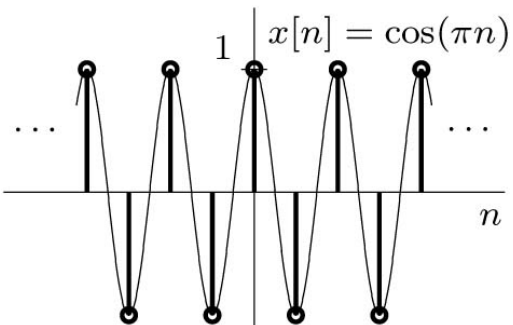
- (a) An upsample by $L = 4$ operation followed by a downsample by $M = 2$ operation.
- (b) A downsample by $M = 2$ operation followed by an upsample by $L = 4$ operation.
- (c) An upsample by $L = 3$ operation followed by a downsample by $M = 2$ operation.
- (d) A downsample by $M = 2$ operation followed by an upsample by $L = 3$ operation.



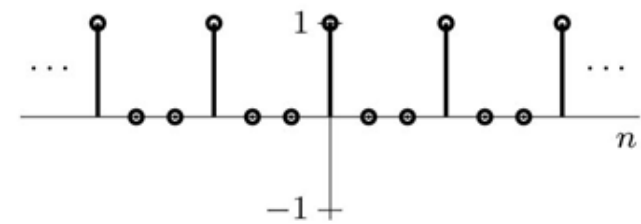
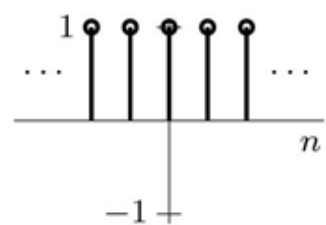
Upsample $L = 4 \rightarrow$ Downsample $M = 2$

Downsample $M = 2 \rightarrow$ Upsample $L = 4$

- (c) An upsample by $L = 3$ operation followed by a downsample by $M = 2$ operation.
- (d) A downsample by $M = 2$ operation followed by an upsample by $L = 3$ operation.



Upsample $L = 3 \rightarrow$ Downsample $M = 2$



Downsample $M = 2 \rightarrow$ Upsample $L = 3$

if L and M are coprime, which is to say they have no common factors other than 1, then the operations of expansion and compression commute. In practice ordering may change aliasing, and speed of operation.

Homework 4 is due: